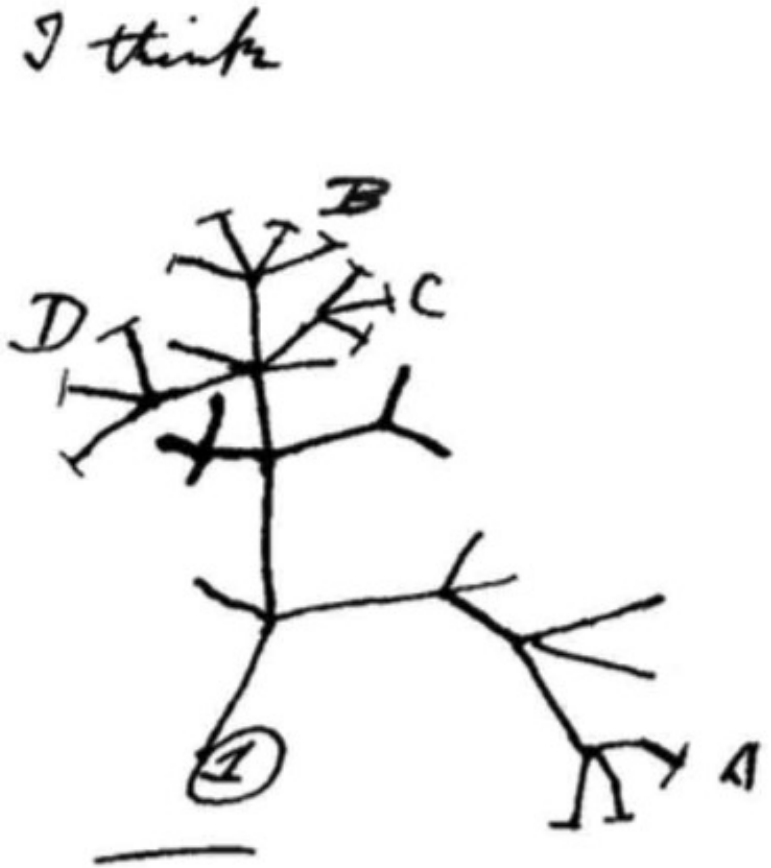
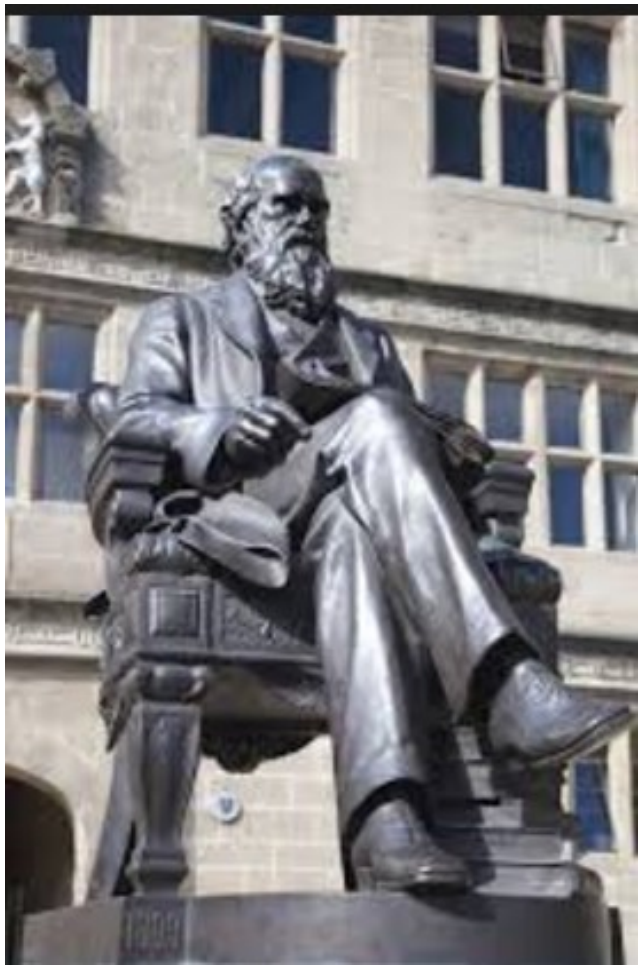
Changes to Allele Frequency in Gene Pools:
The Theory of Evolution Through Natural Selection

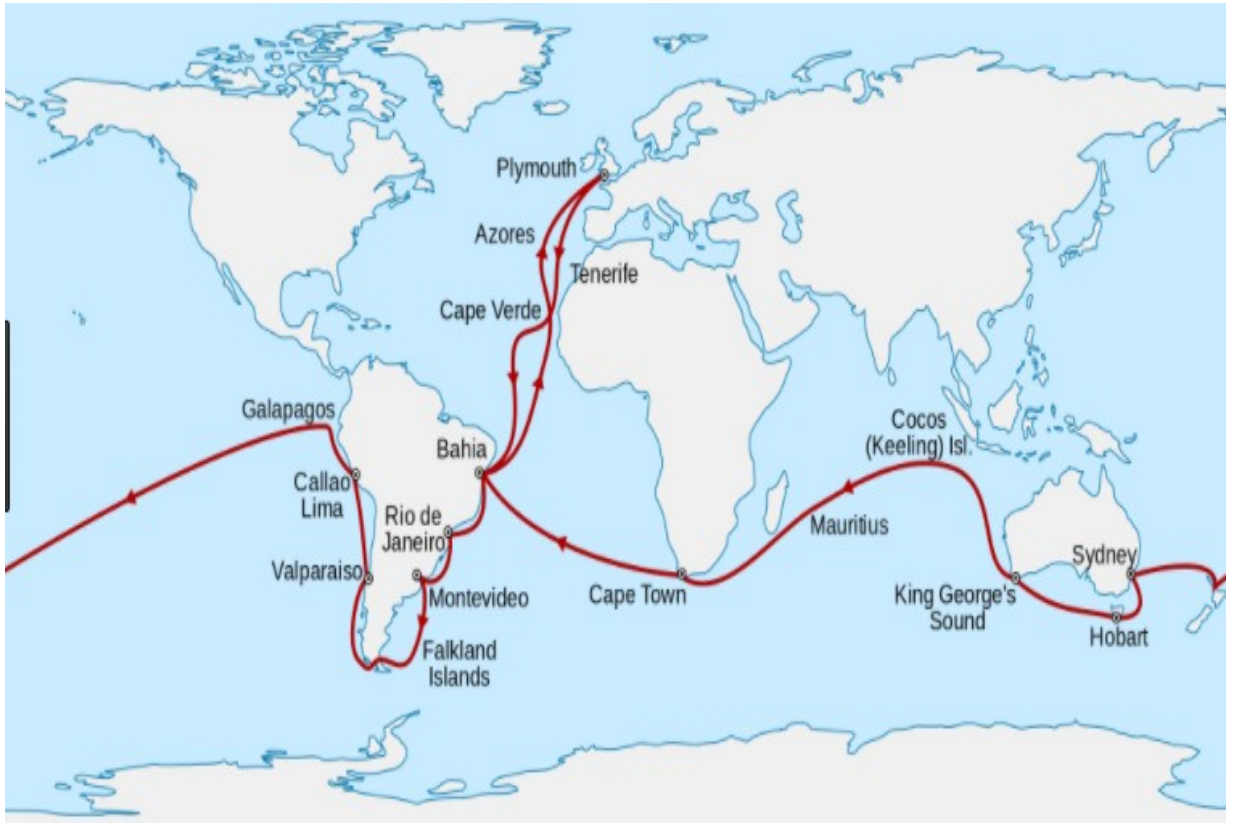




Theory of Evolution Through Natural Selection

- Explanation for the huge diversity of life on Earth
- Underpinned by comprehensive evidence
- Initially put forward in 1858 by:
  - Charles Darwin
  - Alfred Russel Wallace
- Charles Darwin:
  - Keen amateur naturalist
  - Joined a surveying expedition as a biologist
  - Voyaged on HMS *Beagle*
    - Galapagos Islands
    - New Zealand
    - Australia
- Allowed him to observe differences and similarities between geographically separated animals.

The Voyage of the Beagle – 1831-1836



Learning Aim: Outline the historical development of the Theory of Evolution by Natural Selection

Theory of Evolution through Natural Selection

- Built on work of other scientists:
  - Linnaeus – classification
  - Lyell – geological change
  - Malthus – population studies
  - Wallace – similar ideas to Darwin
- Theory based on 3 **observations** made by Darwin.
  - ***Variation***: members of a species vary and variations are passed on from one generation to the next.
  - ***Birth Rate:*** living organisms reproduce at a rate greater than their food supply and must compete for resources
  - ***Nature's Balance:*** although birth rate is high, most species numbers remain at a constant level.

Learning Aim: List the three observations that helped Darwin develop evolutionary theory

Theory of Evolution Through Natural Selection

- Based on the observations, Darwin interpreted and inferred:
    - Excessive birth rate and limited resources → ***struggle for existence***
    - Variations mean that organisms best suited to their environment are more likely to survive → became known as ***"survival of the fittest".***
        - More organisms with favourable characteristics survive
        - Organisms with less favourable characteristics die before they can reproduce
        - Survival of Fittest possible because of variation.

Learning Aim: Describe what Darwin inferred from his observations about Variation, Birthrate and Nature's Balance

Theory of Evolution through Natural Selection

- Darwin did not know ***why*** variation occurs. We now know that a number of things cause this variation across a species, including:
  - Events during meiosis and fertilisation
  - Inheritance of alleles from mother and father for alternative forms of a trait or characteristic
  - Chance mutations

- Today: natural selection can be viewed as *selection of those alleles in a population that give an organism a greater survival advantage*.
  - *Favourable alleles passed to offspring*.
  - *Change in gene pool over time – adaptation of the species to the environment*.

Learning Aim: Give a modern definition of natural selection

Theory of Evolution through Natural Selection

A summary of the principles of evolution through natural selection

1. There is variation of characteristics within a species.
2. More offspring of a species are produced than can possibly survive to maturity.
3. Because of excessive birth rate and limited resources, there is a struggle for existence – competition for survival.
4. Individuals with characteristics best suited to the environment have more chance of surviving and reproducing – survival of the fittest.
5. Favourable characteristics (those with survival value) are passed on to the next generation.
6. In the gene pool, the proportion of alleles that produce favourable characteristics gradually increases.

Learning Aim: Summarise the six principles of Evolution through Natural Selection

Natural Selection is Selective

- Not a random process.

- Allele frequency changes due to environmental factors, making some alleles more favourable for reproduction/survival in a particular environment.

    - Eg: An allele for longer coat length is an advantage in cold climates. Animals expressing this allele are more likely to be able to maintain body temperature to survive and/or produce more offspring, so the allele increases over time in the population



The allele for long coat in the musk ox is an advantage in the arctic climate where it lives. It means the musk ox can survive and reproduce better as it doesn't have to expend as much energy staying warm. Because of this the allele increases in frequency – it is selected for by environmental pressure.

Learning Aim: Discuss how natural selection is SELECTIVE

Natural Selection is Directional

- Frequency of an allele in the population consistently increases over time if it is an advantage, or consistently decreases over time if it is a disadvantage.



There are several variant alleles for darkness of colour in flatfish. Light coloured alleles (and therefore scales) are an advantage in a sandy environment. Fish born with darker scales will be killed more easily by predators so those alleles will decrease in frequency over time, and the alleles for light scales will increase over time as they are an advantage. Provided the environment remains sandy, there will not be an increase in the allele for dark scales.

Learning Aim: Discuss how natural selection is DIRECTIONAL

Natural Selection is Adaptive - influenced by environmental pressures

- change in allele frequency helps the species adapt to its current environment.

- change in allele frequency can clearly be related to environmental factors applying pressure



Mice in New York's Central Park have developed genes to cope with human food. *(Getty: Alan Tunnicliffe)*

Mice in Central Park have a greater allele frequency for alleles that protect against obesity. These alleles are an advantage in an environment where high fat human food waste is easy to come by. High fat foods in the environment apply pressure so that the obesity protection alleles are favoured.

Learning Aim: Discuss how natural selection is ADAPTIVE – influenced by environmental pressures

Genetic Drift vs Natural Selection

Genetic Drift	Natural Selection
Random	Selective
• Allele frequency changes due to random/chance factors eg: ○ A subset of the population becoming isolated ○ Migration and interbreeding between populations ○ More people with a particular allele having more children through random chance • Change of frequency is neither an advantage nor a disadvantage	• Alleles more favourable for survival and reproduction increase as they are passed on to offspring. ○ An allele for longer coat length is an advantage in cold climates. Animals expressing this allele are more likely to survive and/or produce more offspring, so the allele increases over time in the population
Non-directional	Directional
• Frequency of a particular allele can increase and decrease over time due to random factors ○ More red-headed people might randomly have kids in one generation, increasing the allele frequency for red hair, then the next generation they may have less, decreasing the frequency.	• Frequency of an allele in the population consistently increases over time if it is an advantage, or consistently decreases over time if it is a disadvantage.
Non-adaptive	Adaptive
• The change in allele frequency doesn't cause an advantage in the environment	• The change in allele frequency helps the species adapt to its current environment.
Not influenced by environment	Influenced by environmental pressures
• A clear selective agent cannot be identified as being responsible for the change in allele frequency	• A clear selective agent can be identified as responsible for the change in allele frequency.

How can we tell if a change in allele frequency is the result of genetic drift or natural selection?

The key thing to look at is *directionality.* If the gene frequency is consistently increasing over several generations, it is likely because it provides some kind of advantage in the environmental conditions of the population. If the frequency fluctuates up and down over time, it is likely to be random drift. *Note that a permanent drift is more likely where the population is small, even without a selective advantage – eg colourblindness on Pingelap.*

Another thing to look at is a plausible explanation for a consistent, directional change in frequency – *can you link the change to an environmental factor*?

Learning Aim: Compare and Contrast the processes of Genetic Drift and natural selection

Examples of Natural Selection in Humans – Body Stature

- Original gene pool would have alleles for many different body statures (tall, short, stocky, short limbed, etc)



- Environment has selected for certain alleles in different environments:
  - Inuit: long body, short limbs – smaller surface area, less heat loss – adapted to colder conditions



  - Masai: long limbs – larger surface area for heat loss – adapted to warmer climate.

Learning Aim: Give an example of natural selection for traits in the human population

Natural Selection in Action: The Peppered Moth

- Peppered Moth: *Biston betularia*
- Wings originally light mottled – camoflaged against lichen covered tree trunks, as protection from predators.
- Mutation causes dark black wings
- Pre-industrial revolution, 90% of moths had pale wings
- Industrial revolution: coal dust caused tree trunks to be darker.
  - Light coloured moths stood out – eaten by predators
  - Black coloured moths now camouflaged
  - Within 100 years, 90% of moths were black.
  - The coal dust darkening the tree trunks was a *selective agent* for the darker wing allele
- Since the 1950s, pollution control has cleaned the air. Tree trunks are pale again. Moth population now light mottled again! Selection pressures have changed.



Most of the peppered moths collected in the early 1800s were the light form. Click for more detail.



Learning Aim: Describe changes to the Peppered Moths during the industrial revolution as evidence for evolution by natural selection

A great website and game about the peppered moth:



https://askabiologist.asu.edu/peppered-moths-game/play.html

Learning Aim: Describe changes to the Peppered Moths during the industrial revolution as evidence for evolution by natural selection

From the Peppered Moth we can see:

1: There is variation of characteristics within a species – moths carry alleles for light or dark wing colouration

2: More offspring are produced than can survive to maturity – moths are killed by birds, among other things.

3: There is a struggle for existence – moths survival is not guaranteed – a bird might eat them!

4: Individuals with characteristics best suited to the environment have more chance of surviving and reproducing – white moths have more chance of surviving in a birch forest with pale tree trunks.

5: Characteristics with survival value are passed down the to the next generation – the white moths pass the allele for white wing colour on to their offspring at a greater rate than dark moths, because more white moths survive in the pale tree trunk environment.

6: In the gene pool, the proportion of alleles that produce favourable characteristics gradually increases- the allele frequency for white wing colour increases in the pale birch forest.

Natural Selection is Selective- the allele for pale wing colour provides an advantage in the pale trunk birch forest.

Natural Selection is Directional – in the pale trunk forest, the frequency of pale wing colour alleles steadily increases in the direction of advantage. Once the trunk colour changes, the frequency of dark wing alleles rises, again in the direction of the advantage.

Natural Selection is Adaptive and Influenced by Environmental Pressure – tree trunk colour causes environmental pressure that drives adaptation of wing colour – as better camouflaged moths avoid predators better.

Learning Aim: Describe changes to the Peppered Moths during the industrial revolution as evidence for evolution by natural selection

Speciation

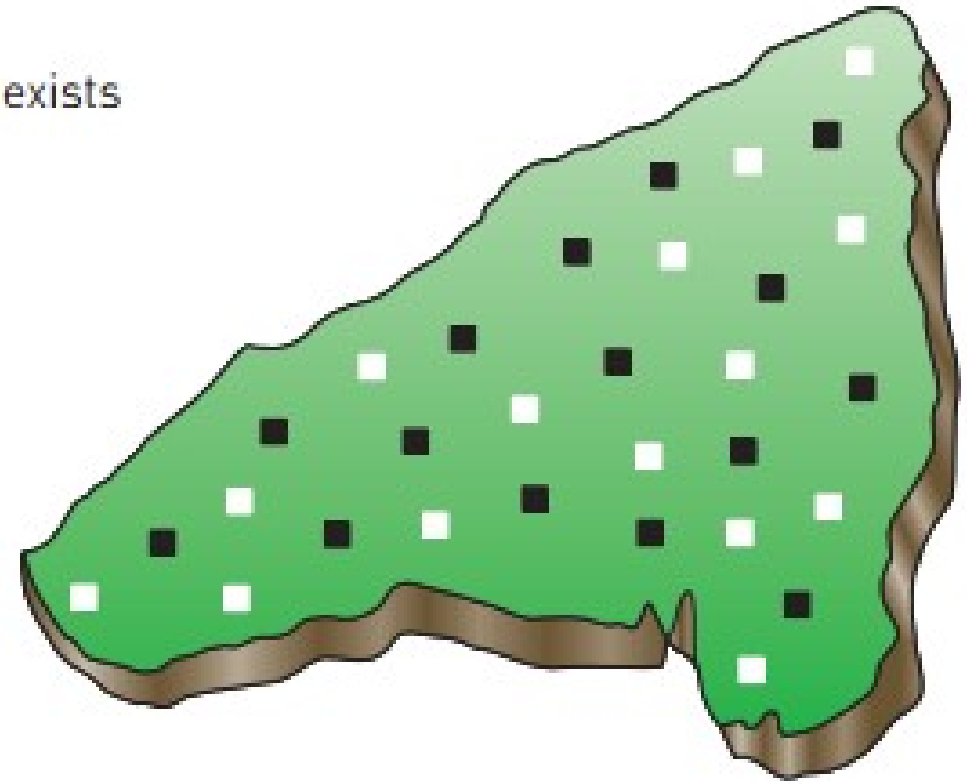
- Splitting of one species into two separate species due to divergence of gene pools.
  - Isolation is a barrier to gene flow

- If two populations are isolated in different environments, different alleles will be favoured – gene pools will diverge over time.

- Over a long period of isolation with different alleles favoured, allele frequencies can differ enough so that reproduction is no longer possible between populations: the two populations have become ***separate species***.

Learning Aim: Define Speciation

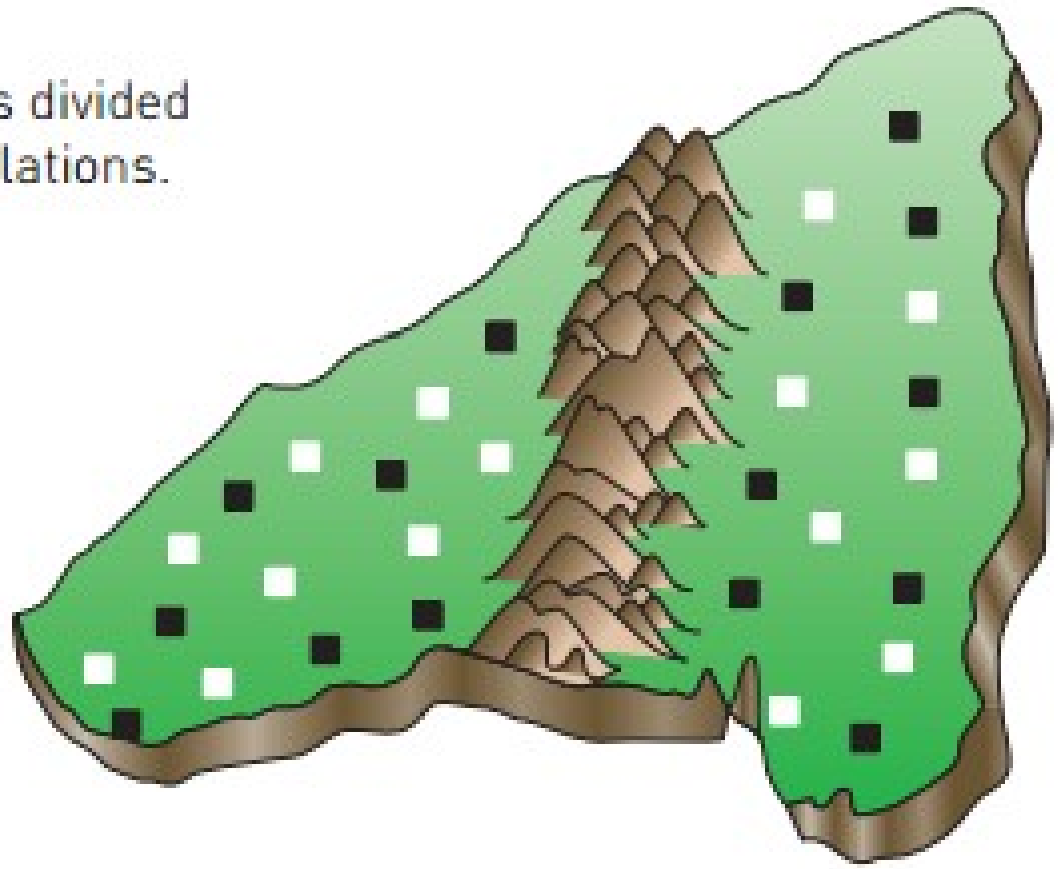
a Variation
A population exists on an island.



A range of **variations** exists within the population, which shares a **common gene pool**.

Learning Aim: List and describe the processes involved in speciation

b Isolation
The species is divided into two populations.

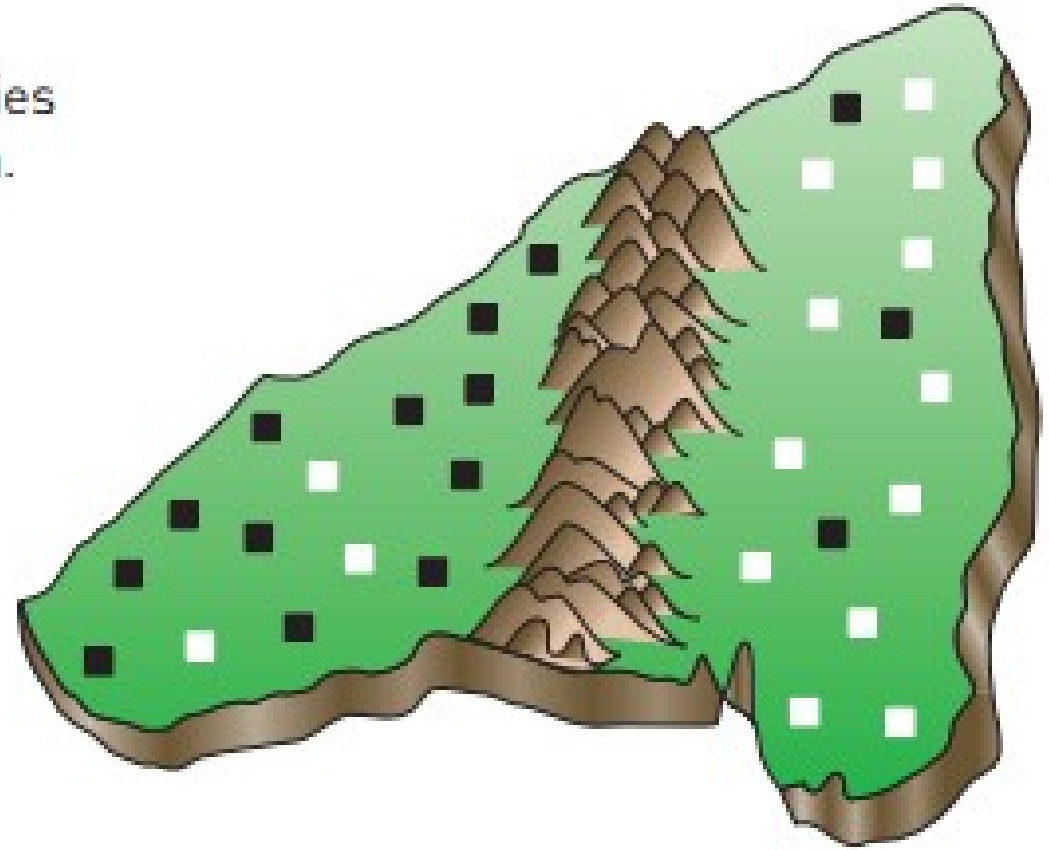


A **barrier** has formed, dividing the population into two. No interbreeding occurs between the two populations. Each population has a **separate gene pool**.

**note – a physical barrier is not always necessary for speciation. If a species is spread over a wide geographic area with different environmental pressures, distance becomes the barrier to reproduction, and selection will take place to suit each environment, even if there is some genetic drift.*

Learning Aim: List and describe the processes involved in speciation

c Selection
Two subspecies begin to form.

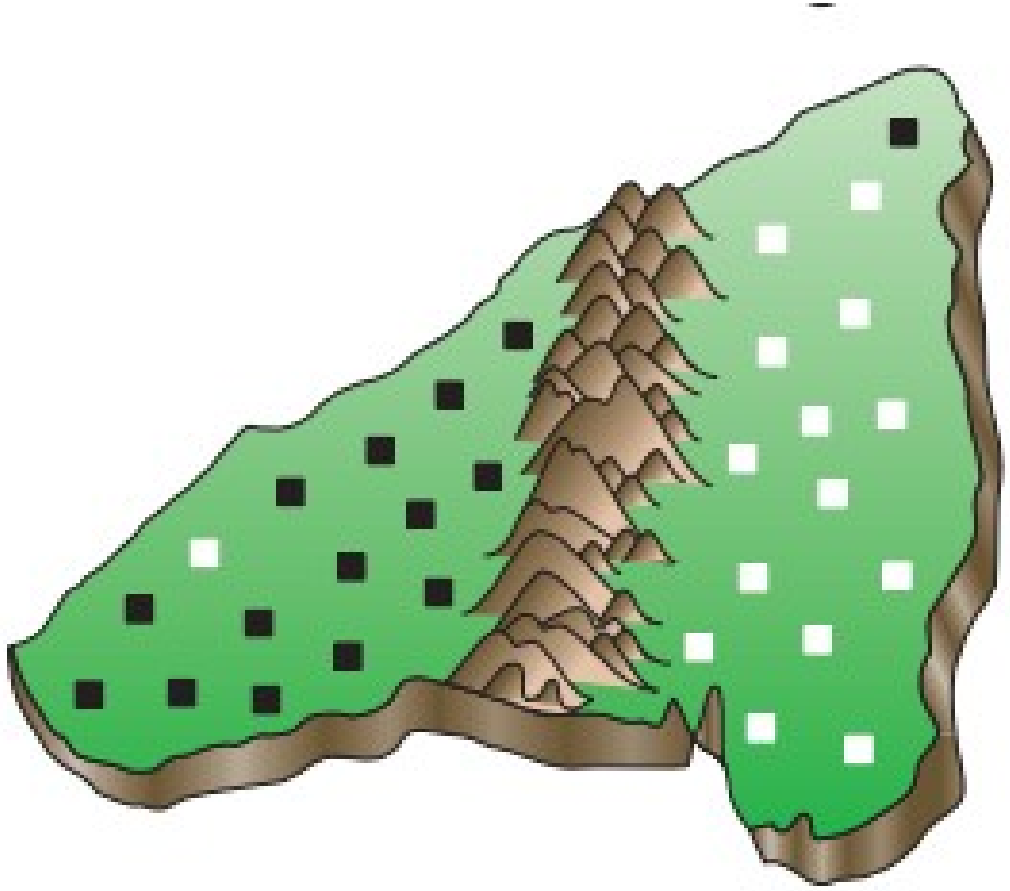


Different **selection pressures** act on each of the two populations over a number of generations. This brings about a change in the **gene frequencies** of each gene pool. Such changes lead to the evolution of separate **subspecies**.

Learning Aim: List and describe the processes involved in speciation

d Speciation

Two species now exist.



Over a long period of time the changes in the gene frequencies may be great enough to prevent the production of fertile offspring by interbreeding between the two populations from ever occurring again. When this happens, *two* **species** exist.

Learning Aim: List and describe the processes involved in speciation

Galapagos Finches and Speciation

- Galapagos Islands –
  - Group of volcanic islands in ocean off South America.
  - Environment varies widely between islands due to
    - Age of island – younger islands – less vegetation
    - Ocean currents and mountains affecting weather

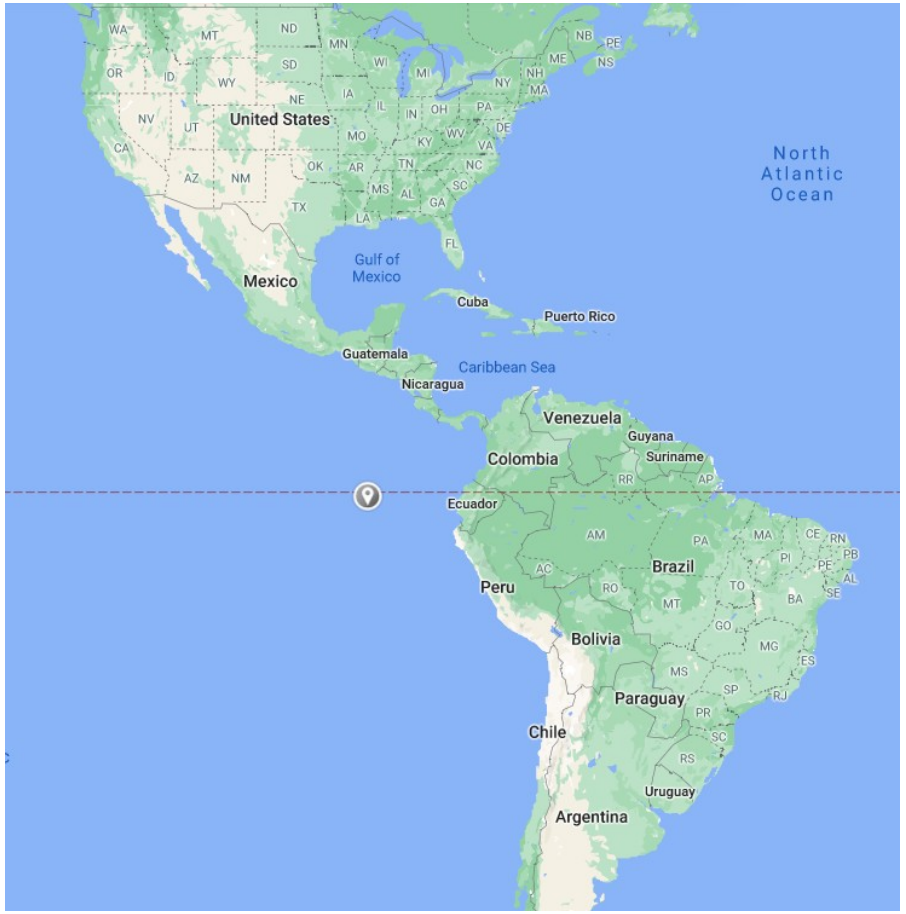


Pinzon Island – arid environment



Humid environment on Santa Cruz Island

- Different environments mean different food sources available

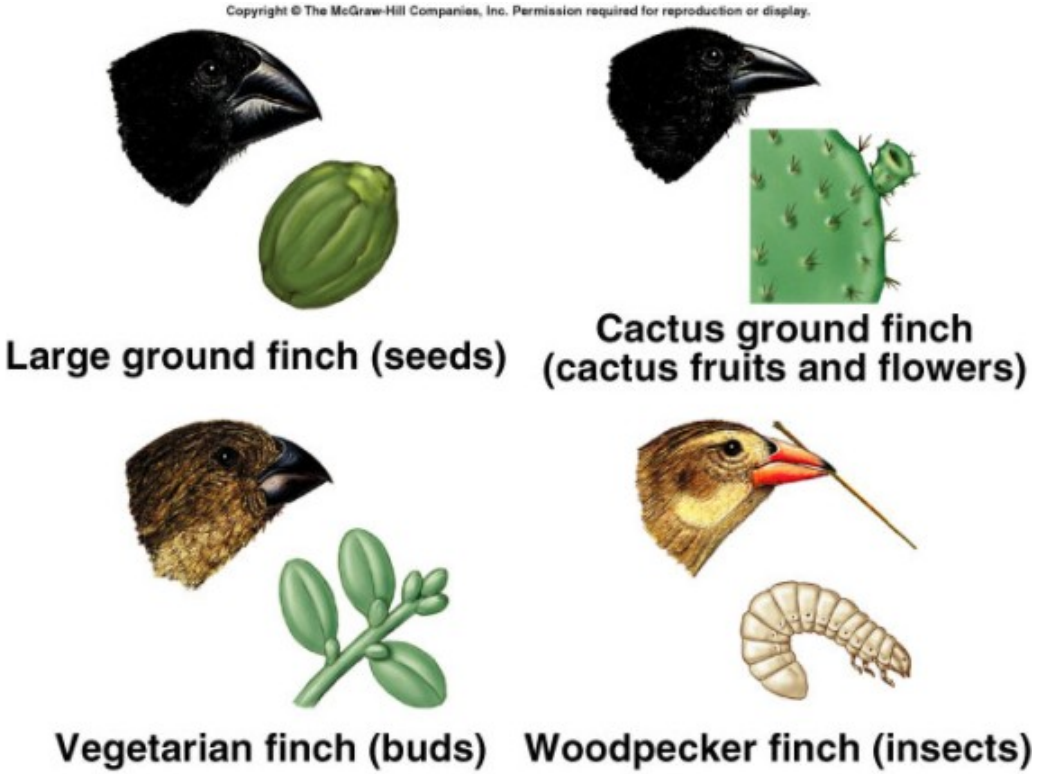




Learning Aim: Discuss how speciation occurred for finches on the Galapagos Islands

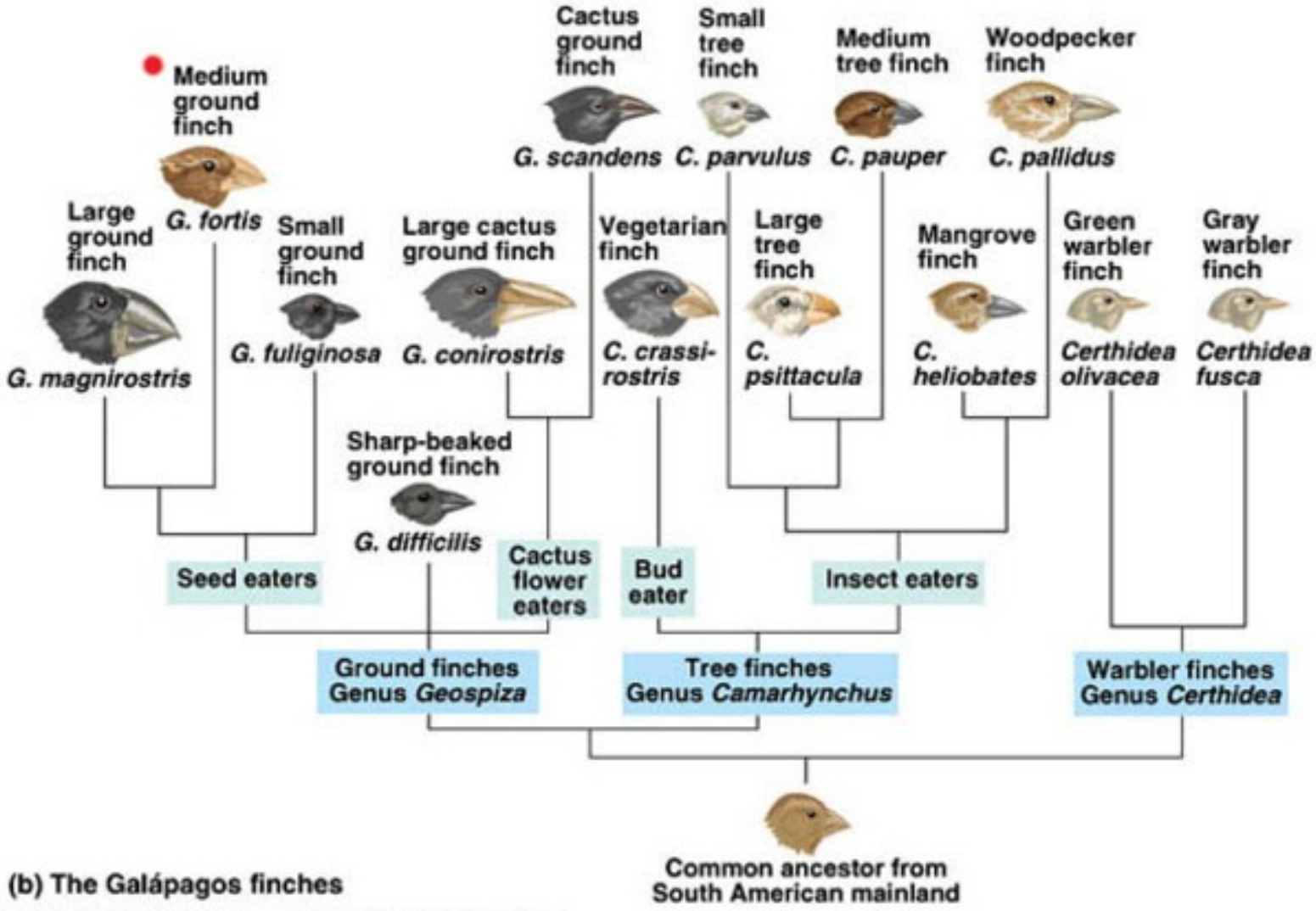
Galapagos Finches and Speciation

In the distant past, finches from the South American mainland arrived in the Galapagos, most likely blown there during storms.
Mainland too far away for gene flow between populations.
Finches colonised different islands, with different climates and food sources.
Limited gene flow between islands
Finch species are adapted (by natural selection) to the food sources on the islands on which they are found.



Learning Aim: Discuss how speciation occurred for finches on the Galapagos Islands

Galapagos Finches and Speciation



Learning Aim: Discuss how speciation occurred for finches on the Galapagos Islands